
Session 1: Introduction to Computer Forensics

- Computer Forensics Overview
 - Computer Crime vs Unauthorized Activities
 - Cyber Laws
 - Six-Step Forensic Process
 - Need for a Forensic Investigator
-

Session 2: Fundamentals of Cyber Forensics

- Computer Forensics Process
 - Preservation
 - Identification
 - Extraction
 - Documentation
 - Interpretation
 - Goals of Forensic Analysis
 - Types of Cyber Forensics
 - Cyber Forensics Procedures
 - Incident Preparation
 - Pre-Incident Activities
 - Incident Response Plan
 - Incident Response Team (IRT)
 - Incident Detection
 - Chain of Custody
-

Session 3: Evidence Handling

- Evidence Checkout Log
- Evidence Handling
- First Response
- Response Strategy
- Forensic Duplication (Imaging)
- Evidence Authentication
- Investigation Process
- Common Investigation Mistakes
- Detection Techniques

Lab

- Evidence Checkout Log
 - Investigation
 - Detection
-

Session 4: Initial Assessment & Hexadecimal

- Initial Assessment
 - Incident Notification Checklist
 - Hexadecimal Number System
 - Practical Bits
 - Binary to Hex Conversion
 - Applications of Hexadecimal
-

Session 5: Encoding, Encryption & Hashing

- Encoding
- Encryption
- Hex Editor
- File Structure
- Hashing Concepts
- Hashing DLLs

- MD5
- Hash Collisions
- Bit Rot

Lab

- Encoding
 - Encryption
-

Session 6: Crime Scene Processing

- Standard Operating Procedures (SOP)
 - Processing Crime & Incident Scenes
 - Windows Forensics
 - DOS Commands for Forensics
-

Session 7: Investigation & Legal Aspects

- Forensic Implications
 - Accreditation Standards
 - Cyber Forensics Investigation
 - Privacy in Cyber Forensics
-

Session 8: Computer Forensics Tools

Theory

- Sysinternals Suite
- FTK Toolkit
- FTK Imager
- OSForensics (OSF)
- Hex Editors (HxD)

Lab

- Cyber Check Suite

- Practical using forensic tools
-

Session 9: Live & Mobile Forensics

Theory

- Live System Forensics (Active State)
- Windows Live Forensics
- Linux Forensics
- Linux Artefact Collection
- Mobile Forensics Basics

Lab

- Live System Forensics
 - Linux Forensics Practical
-

Important Practical Tools

- Sysinternals Suite
 - FTK Imager
 - FTK Toolkit
 - OSForensics
 - HxD Hex Editor
 - Autopsy (Recommended)
 - Volatility (Memory Analysis)
 - LiME (Linux Memory Capture)
-

Important Practical Topics

- Disk Imaging
- Hash Verification (MD5, SHA-1, SHA-256)
- Metadata Analysis
- Hex Analysis
- String Search
- Deleted File Recovery
- Live Memory Capture

- Process Analysis
- Network Connections
- Linux Log Analysis
- Mobile Data Acquisition
- Chain of Custody Documentation